

01 / BEYOND THE SWIPE / THE GREENER CHECKOUT

THE HIDDEN RECEIPT

Every payment leaves a financial receipt - and an environmental one.

We only see the first. The second is spread across plastic, paper, terminals, data centers and settlement layers. SlicePay exists to make new finance feel simpler at checkout and more accountable in its footprint.

1/30

END-TO-END DESIGN TARGET

vs. Visa operational energy
allocation per transaction

One payment. Two outcomes. Value moved. Impact measured.

SlicePay turns QR, POS and web checkout into stablecoin settlement, loyalty and staff rewards - without asking the merchant or customer to become a blockchain operator.

0.00412 Wh

SOLANA NETWORK / TX

Underlying rail estimate
before SlicePay application load

Directional research brief. Product-level lifecycle assessment is proposed, not yet completed.



02 / ONE PURCHASE, TWO SYSTEMS

A \$25 payment is never just \$25.

The customer sees a coffee, a meal or a service. The merchant sees margin and settlement. The infrastructure sees credentials, devices, processors, networks, ledgers and energy. SlicePay begins where all three realities meet.

THE HUMAN THESIS

The card system made payment invisible. New finance must make impact visible.

SlicePay packages efficient multi-chain settlement into QR, POS and hosted web checkout, while the merchant receives USDC or USDT without changing how the team serves a customer.

2.45 g

CARD LIFECYCLE

0.74 g

OPTIMIZED DIGITAL

0.99 Wh

VISA OPERATING ALLOCATION

1.9%

SLICEPAY TOTAL FEE

TRADFI: TRUST AT GLOBAL SCALE

It solved acceptance and protection.

Banks and card networks deliver ubiquity, familiar consumer protection, dispute processes, fraud controls and regulated access. Visa also purchases 100% renewable electricity for its offices and data centers.

The point is not that TradFi ignores climate. It is that the system was designed before per-payment environmental accountability was a product requirement.

THE STRUCTURAL MEASUREMENT GAP

Banks report portfolios, not receipts.

PCAF and GHG Protocol methods help banks measure financed emissions from loans and investments. Corporate reports cover operations and suppliers. Those are essential views, but neither is a standard carbon receipt for one completed payment.

Programmable rails create the raw material for that missing layer: timestamped routes, action counts and settlement records.

WHY SLICEPAY EXISTS

01 Abstract the technology

Stablecoin supply crossed about \$389B and adjusted 12-month volume exceeded \$8.1T in 2026, yet less than 1% was small-ticket retail. People do not need another chain; merchants need usable acceptance.

02 Unify the merchant outcome

One layer connects wallet payment, USDC or USDT settlement, loyalty and staff incentives at a transparent 1.9% total fee.

03 Turn accountability into infrastructure

Telemetry today can become verified CO2e tomorrow: a new payment receipt that supports merchant reporting, enterprise procurement and customer trust.

TradFi carries a lifecycle. Efficient rails carry less.

TRADFI LIFECYCLE CARBON

2.45 g

CO2E / IN-STORE DIGITAL PAYMENT



- Receipt 1.03 g
- Physical card 0.84 g
- Terminal 0.49 g
- Data centers 0.09 g

70% lower: Worldline models 0.74 g when the receipt, physical card and dedicated terminal are removed through phone-to-phone design.

OPERATING ENERGY / TRANSACTION

Efficient onchain rails start lighter.

Wh - LOG SCALE

VISA OPERATING ALLOCATION

0.993 Wh

STELLAR NETWORK ESTIMATE

0.320 Wh

SLICEPAY DESIGN BUDGET

<=0.033 Wh

SOLANA NETWORK RAIL

0.00412 Wh

~241x

lower: Solana network vs. Visa operational allocation

PAYMENT ARCHITECTURE MATRIX - SIMPLE VIEW



Directional positioning: SlicePay remains a target until product telemetry and lifecycle assessment.

THE SLICEPAY EQUATION

$$E(\text{payment}) = E(\text{checkout} + \text{cloud}) + n \times E(\text{route}) + E(\text{settlement} + \text{rewards})$$

Design budget: less than or equal to 0.033 Wh per completed payment, equal to 1/30 of the FY2025 Visa operating-energy allocation calculated from 255,688 MWh divided by 257.5B processed transactions. One Solana network transaction uses 0.00412 Wh, leaving about 8.0x total headroom. Four onchain actions would consume about 0.0165 Wh before SlicePay application and device load.

SlicePay = efficient rail + merchant abstraction + evidence.

The value is not crypto at checkout. It is better infrastructure underneath.

Boundary rule: never compare network-only Wh with full-lifecycle CO2e. Telemetry plus LCA connects the two.

04 / THE MISSING MERCHANT LAYER

Frictionless for the merchant. Measurable by design.

\$389B

GLOBAL STABLECOIN SUPPLY / 2026

\$8.1T+

ADJUSTED 12-MONTH VOLUME

<1%

SMALL-TICKET RETAIL SHARE

ONE FLOW, LESS OPERATIONAL FRICTION

CUSTOMER PAYS

Any compatible wallet through QR, POS or hosted web checkout.

SLICEPAY ORCHESTRATES

Routes payment, settlement, loyalty and optional staff incentive.

MERCHANT RECEIVES

USDC or USDT with transparent 1.9% pricing and one workflow.

ILLUSTRATIVE MERCHANT COST COMPARISON

\$250K annual volume / \$50 average ticket / 5,000 successful payments

Gross processing fees only. U.S. public list-price scenario; not a universal merchant quote.

ONLINE DOMESTIC CARD



\$8,750

IN-PERSON DOMESTIC CARD



\$7,000

SLICEPAY



\$4,750

\$2,250-\$4,000

GROSS ANNUAL SAVINGS
versus the two card scenarios

32%-46%

LOWER PROCESSING COST
before conversion or off-ramp costs

0.3%

STAFF COMMISSION
included within the 1.9% fee

THE SLICECHAIN MISSION IN ONE LINE

Make new finance feel simpler than old finance - and measurable enough to earn trust.

SlicePay is the merchant-facing layer: one acceptance experience, programmable settlement and accountable impact.

05 / FROM MISSION TO EVIDENCE

The greenest payment proves itself.

SliceChain's mission is simple: finance that is easier for people, better for merchants and accountable by design. Measurement comes first; the claim follows.

RECOMMENDED CLAIM LADDER

NOW

Designed toward $\leq 1/30$

Use as an engineering target versus Visa operational energy allocation. State that product LCA is pending.

WITH TELEMETRY

Measured Wh / payment

Publish route mix, action count, retries, checkout and cloud energy with P50 and P95 results.

AFTER ASSURANCE

Verified CO₂e / payment

Only then make comparative lifecycle claims and issue merchant-level green checkout reports.

HIGH-IQ OPERATING RECOMMENDATIONS

BUILD THE EVIDENCE ENGINE

- 01** Instrument every completed and failed payment. Count route, network actions, retry loops, cloud load and checkout-device time.
- 02** Optimize total impact, not chain energy alone. Security, liquidity, compliance, finality and failure rate can outweigh a small Wh difference.
- 03** Report distributions. P50 and P95 reveal expensive edge cases hidden by a simple average.
- 04** Commission an independent LCA. Use one functional unit: a completed merchant payment, end to end.

POSITION THE COMMERCIAL WEDGE

- 01** Lead with merchant outcomes: lower gross fees, one checkout layer, stablecoin settlement, loyalty and staff incentives.
- 02** Do not claim universal price leadership. Stripe currently advertises 1.5% for stablecoin payments; SlicePay's 1.9% must win on bundled operating value.
- 03** Run a 90-day merchant cohort. Compare SlicePay with actual processor statements, settlement time, failures and repeat purchase behavior.
- 04** Publish a Green Receipt only after verification. Show method version, system boundary and route mix with every estimate.

ENVIRONMENTAL ECONOMICS - KEEP IT HONEST

~\$129

ENERGY SPEND / 1M PAYMENTS

Illustrative operational delta: 0.993 Wh less 0.033 Wh, valued at the 2025 U.S. commercial electricity average of 13.41 cents/kWh.

These are societal and energy values, not merchant cash savings. Fee and workflow economics carry the P&L case.

~\$325

AVOIDED SOCIAL DAMAGE / 1M

If an LCA verifies 0.74 g versus the 2.45 g digital-payment baseline: 1.71 tCO₂e avoided at an illustrative \$190/t EPA social cost.

RESEARCH SOURCES - CURRENT THROUGH AUGUST 2026

- [1] SlicePay product, 1.9% fee and merchant flow
- [2] Worldline 2025 sustainability report and payment LCA
- [3] Visa 2025 sustainability report: 255,688 MWh
- [4] Visa FY2025 annual report: 257.5B processed transactions
- [5] Solana September 2024 energy impact report
- [6] Ethereum proof-of-stake energy consumption
- [7] Stellar and PwC blockchain sustainability framework

- [8] Visa 2026 stablecoin scale and retail-payment gap
- [9] Stripe current card and stablecoin pricing
- [10] EIA 2025 U.S. commercial electricity price
- [11] EPA 2023 social cost of greenhouse gases report
- [12] DNB / Maastricht debit-card lifecycle assessment
- [13] Visa July 2026 stablecoins as money-movement rail
- [14] GHG Protocol Product Life Cycle Standard
- [15] PCAF / GHG Protocol financial-industry standard